# Location

- CPU
  - Registers, Control Memory
- Internal
  - Cache
  - Main Memory – RAM
  - ROM
- External
  - Disk, Tape, ...

# Memory

- *Computer memory is the storage space in the computer, where data and instructions required for processing are stored.*
- The memory is divided into large number of small parts called blocks or locations.
- Each location or block has a unique address, which varies from zero to memory size minus one.
- Classifications: Internal and External Memory

Generative summary

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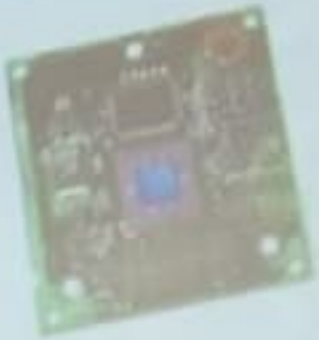
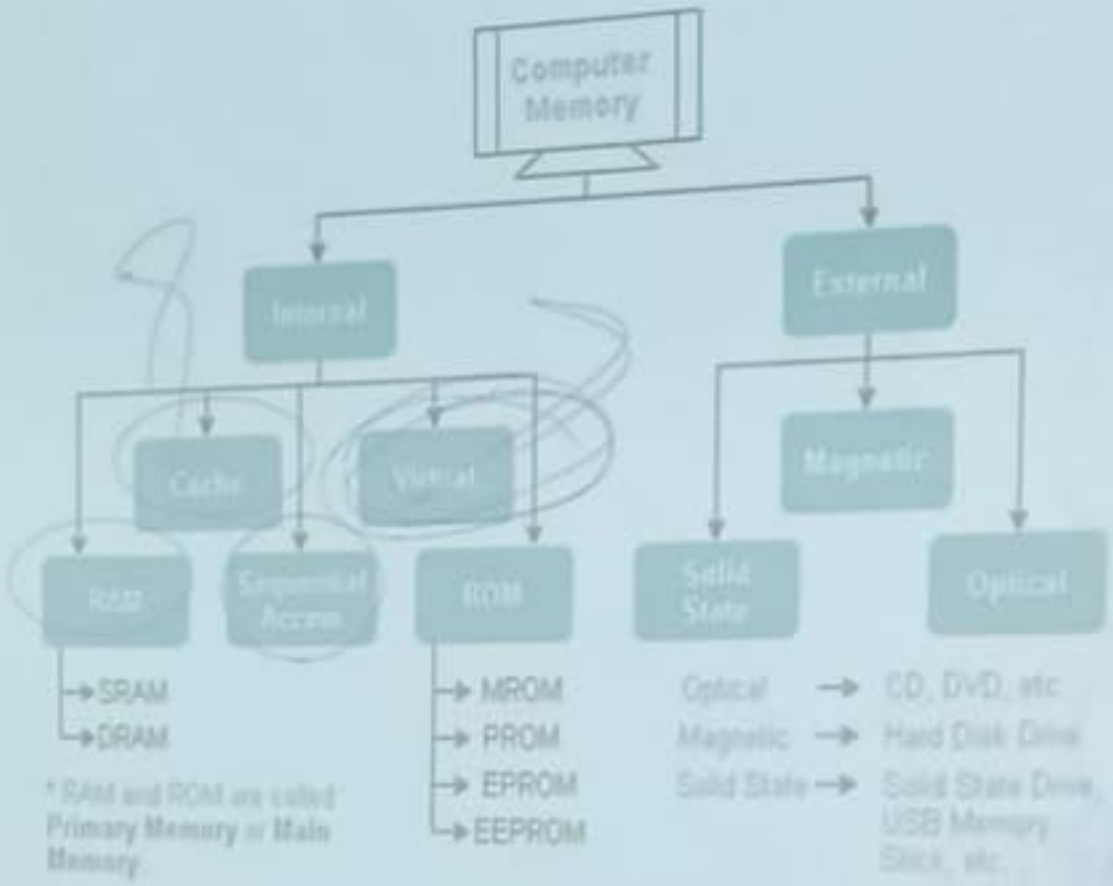
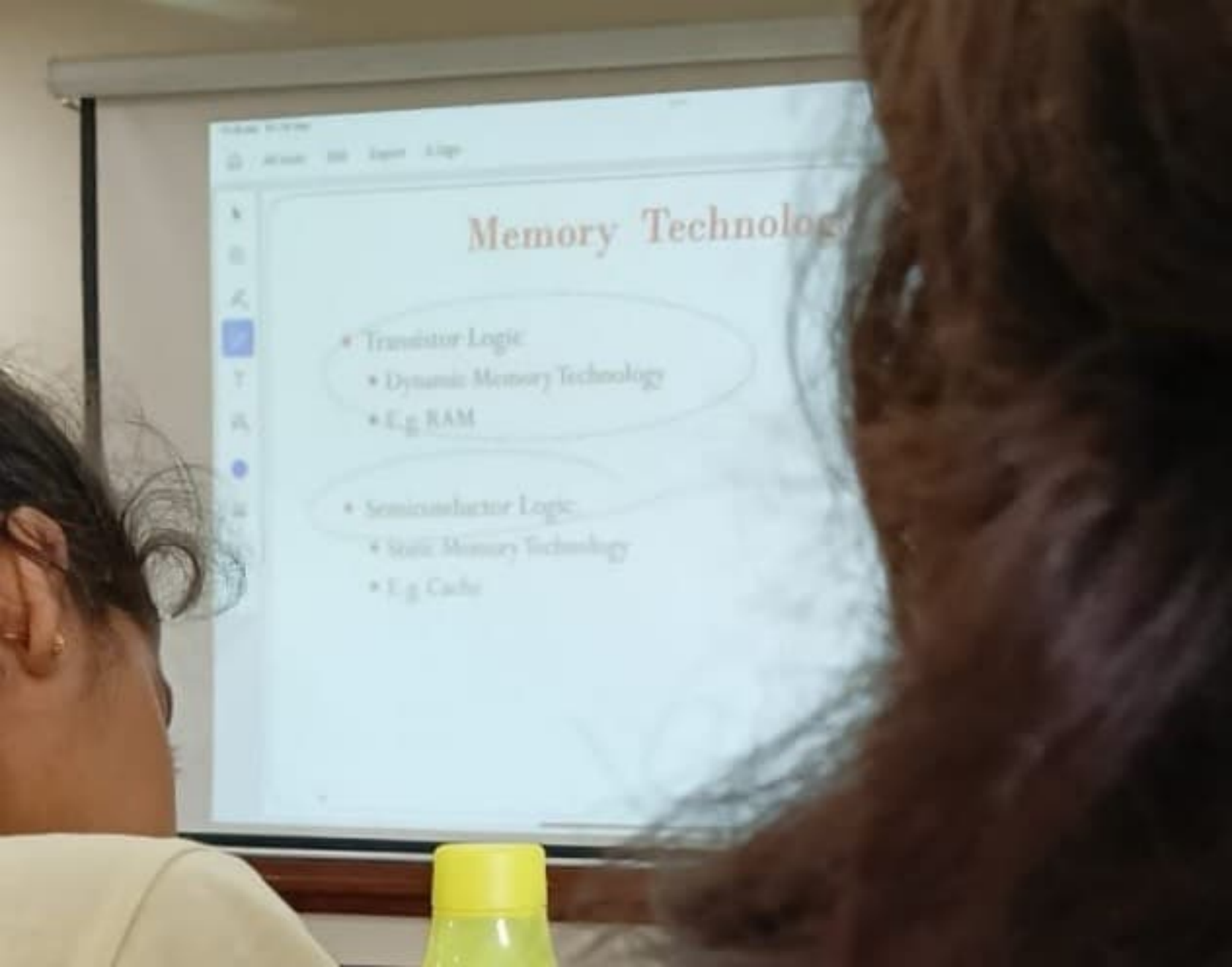


# Memory Classification

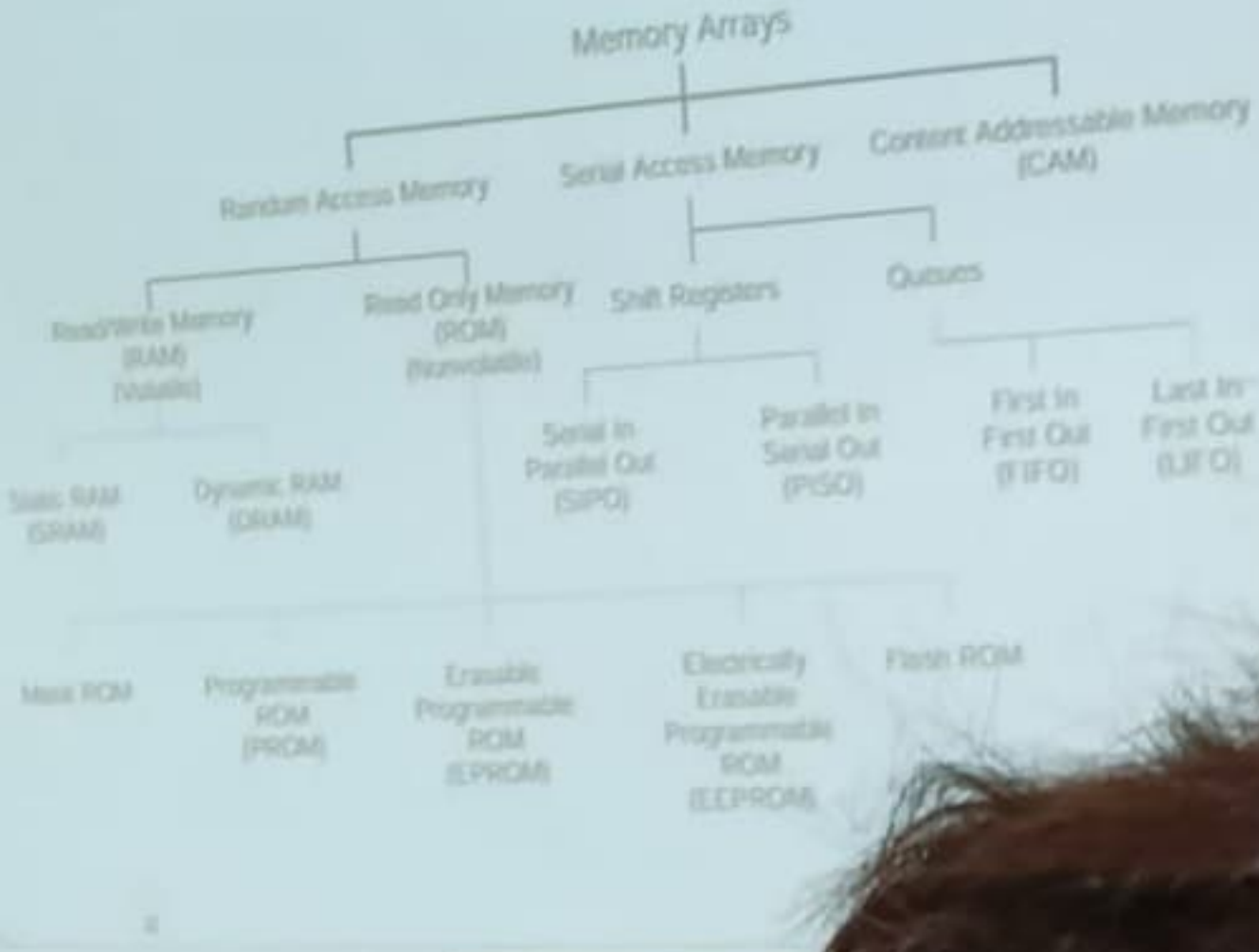



Memory Technolog
Transistor Logic
Dynamic Memory Technology
E.g RAM
Semiconductor Logic
Static Memory Technology
E.g Cache

# Key Aspects of Semiconductor Technology

- Organization
- RAM
- ROM
- Chip Logic
- Chip Packaging
- Module Organization
- Interleaved Memory

# Internal Memory Arrays



# Random Access Memory

- The most common semiconductor memory is referred to as random-access memory (RAM).
- One distinguishing characteristic of RAM is that it is possible both to read data from the memory and to write new data into the memory easily and rapidly.
- Both the reading and writing are accomplished through the use of electrical signals.
- Another distinguishing characteristic is that it is volatile.
- A RAM must be provided with a constant power supply.
- If the power is interrupted, then the data are lost.
- RAM can be used only as temporary storage.

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# RAM Properties

- Misnamed as all semiconductor memory is random acc
- Read/Write
- Volatile
- Temporary storage
- Static or dynamic

# SRAM

- A static RAM (SRAM) is a digital device that uses the same logi
  elements used in the processor.
- In a SRAM, binary values are stored using traditional flip-flo
  logic-gate configurations
- A static RAM will hold its data as long as power is supplied

# SRAM Design

TI 65nm:
0.46x1.06um²

IBM 65nm:
0.41x1.25um²

Intel 65nm:
0.46x1.24um²

# DRAM

- DRAM is made with cells that store data as charge on capacitors.
- The presence or absence of charge in a capacitor is interpreted as a binary 1 or 0.
- Capacitors have a natural tendency to discharge.
- Dynamic RAMs require periodic charge refreshing to maintain data storage.
- *The term dynamic refers to this tendency of the stored charge to leak away, even with power continuously applied.*

# DDRAM

- Computers built before 2002 generally used SDRAM.
- Memory technology took a revolutionary stride with Double Data Rate (DDR) option.
- Current computers use DDR RAM.
- Since computer hardware is all connected and interdependent, this can lead to speed increases in other components, too.
- So, upgrading computer memory is a great way to fix a slow computer.

# DDRAM Generations

DDR

DDR2

DDR3

DDR4

# DDRAM

| | SDRAM | DDR |
|---|---|---|
| Prefetch | 1 - Bit | 2 - Bit |
| Data Rate (MT/s) | 100 - 166 | 266 - 400 |
| Transfer Rate (GB/s) | 0.8 - 1.3 | 2.1 - 3.2 |
| Voltage (V) | 3.3 | 2.5 - 2.6 |

# DDRAM 2

- *DDR2 was introduced in 2003 and operates twice as fast as DDR due to an improved bus signal.*
- DDR2 uses the same internal clock speed as DDR, however, the transfer rates are faster due to the enhanced input/output bus signal.
- *DDR2 has a 4-bit prefetch, which is twice that of DDR.*
- DDR2 can also reach data rates of 533 to 800MT/s.
- *DDR2 memory can be installed in pairs to run in "dual channel mode," which can increase memory throughput even more.*

# DDRAM 2

| | DDR | DDR2 |
|---|---|---|
| Prefetch | 2 - Bit | 4 - Bit |
| Data Rate (MT/s) | 266 - 400 | 533 - 800 |
| Transfer Rate (GB/s) | 2.1 - 3.2 | 4.2 - 6.4 |
| Voltage (V) | 2.5 - 2.6 | 1.8 |

# DDRAM 4

| | DDR3 | DDR4 |
|---|---|---|
| Prefetch | 8 - Bit | Bit per Bank |
| Data Rate (MT/s) | 1066 - 1600 | 2133 - 5100 |
| Transfer Rate (GB/s) | 8.5 - 14.9 | 17 - 25.6 |
| Voltage (V) | 1.35 - 1.5 | 1.2 |

# DDRAM 5

| | DDR4 | DDR5 |
|---|---|---|
| Prefetch | Bit per Bank | 16 - Bit |
| Data Rate (MT/s) | 2133 - 5100 | 3200 - 6400 |
| Transfer Rate (GB/s) | 17 - 25.6 | 38.4 - 51.2 |
| Voltage (V) | 1.2 | 1.1 |

# DDR Comparisons

| | DRAM | DDR | DDR2 | DDR3 | DDR4 | DDR5 |
|---|---|---|---|---|---|---|
| Prefetch | 1-Bit | 2-Bit | 4-Bit | 8-Bit | Bit per Bank | 16-Bit |
| Data Rate (MT/s) | 100-166 | 266-400 | 533-800 | 1066-1600 | 2133-5100 | 3200-6400 |
| Transfer Rate (GB/s) | 0.8-1.3 | 2.1-3.2 | 4.2-6.4 | 8.5-14.9 | 17-25.6 | 38.4-51.2 |
| Voltage (V) | 3.3 | 2.5-2.6 | 1.8 | 1.35-1.5 | 1.2 | 1.1 |